PHYS400
SPECIAL PROBLEMS IN PHYSICS
INTERIM REPORT

Project Title	Automated Construction of Multi-Element MEAM Interatomic Potentials via DFT-Informed Neural Network Optimization
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Abstract

Molecular dynamics (MD) simulations of metallic alloys require interatomic potentials that describe the interactions between every constituent element pair. The Modified Embedded Atom Method (MEAM) provides such a framework, but published MEAM parameter sets typically cover only narrow subsets of the periodic table. When the elements of an alloy are not present in a single potential file simulation cannot be executed. This project develops an automated pipeline that generates a MEAM parameter file for a selebtable list of elements by the following steps: (i) randomized alloy configurations from the readily available configuration files, (ii) LAMMPS evaluation of Young's modulus and Poisson's ratio, (iii) density functional theory (DFT) reference data computation in Quantum Espresso through the Atomic Simulation Environment, and (iv) DFT-driven MEAM parameter initialization with automatic merging of existing libraries. A divergent, composition-only neural-network surrogate (mapping element fractions directly to E and ν) has also been trained as a fast sanity check on the consistency of the LAMMPS dataset. The neural-network parameter optimization that closes the loop on top of Stage 4 and its formal validation against industrial alloys are planned for the second half of the semester.

Introduction

Motivation

Industrial alloys, such as steels and aluminium alloys, are selected for mechanical components on the basis of their elastic properties: the Young's modulus E controls stiffness while the Poisson's ratio ν controls the transverse response under load. Today, these properties are obtained from experiments, which are both expensive and time-consuming. If a sufficiently accurate and inexpensive computational alternative existed, it would accelerate the development of new alloy compositions for specific mechanical demands.

Molecular dynamics (MD) simulations are the natural candidate. The reliability of an MD prediction is, however, set entirely by the interatomic potential it utilizes. Among the available functional forms, the Modified Embedded Atom Method (Baskes, 1992; Lee and Baskes, 2001) is the de-facto standard for metallic and covalent systems. It is computationally cheap relative to machine-learned potentials, analytic, and physically interpretable.

The Cross-Interaction Problem

The fundamental obstacle to applying MEAM to industrial alloys is that published MEAM parameter sets cover a limited subset of elements. A typical paper provides a 3-4 element set, for example the Al–Mg–Zn potential of Dickel et al. (2018) or the Al–Si–Mg–Cu–Fe set of Jelinek et al. (2012), and is silent on cross-interactions with elements that lie outside its scope. A practically relevant alloy such as Al 7075 contains Al, Zn, Mg, Cu, Cr, Mn, Fe, Si and Ti, and no single published MEAM file spans this composition. Naively concatenating parameters from different libraries leaves the binary cross-terms undefined and may produce unphysical or unstable simulations.

Two routes around this problem have been pursued in the literature. The first is manual fitting of new MEAM parameters against experimental and reference databases (Jelinek et al., 2012), which is slow and does not generalize to new element sets. The second is to abandon MEAM in favour of fully data-driven machine-learning interatomic potentials (Behler and Parrinello, 2007; Shapeev, 2016; Mishin, 2021); these are flexible but require large training sets and lose the analytic readability that makes MEAM useful in industry.

Approach Adopted in This Project

The strategy adopted in this project is hybrid combination of the mentioned methods. Rather than replacing MEAM, neural networks will eventually be used as *surrogates* that optimize the MEAM parameters themselves, starting from values seeded by DFT calculations and trying to achieve the calculated E and ν values from the available MEAM files. This preserves the computational efficiency and transparency of the MEAM formalism while exploiting DFT accuracy to fill the missing cross-interaction terms. The four stages of the pipeline that are in scope for this interim report—configuration generation, MD evaluation as baseline, DFT reference calculations, and MEAM initialization—are described in detail in Section 2; a divergent, composition-only neural-network surrogate, used as a sanity check on the LAMMPS dataset, is described in Section 2.6. The neural-network parameter optimization that will close the loop is left to the second half of the semester.

Research Question

Can DFT reference calculations combined with neural-network surrogate optimization produce multi-element MEAM interatomic potentials that accurately reproduce the elastic properties of arbitrary metallic alloys, even when the constituent elements do not share a common published MEAM parameterization?

Project Progress

Pipeline Architecture

The project is implemented as several Python modules that will all be combined in a pipeline of five sequential stages. Stages 1–4 are complete and described in this interim report; Stage 5 (the neural-network parameter optimization that closes the loop on top of Stage 4) is planned for the second half of the semester. Each stage is also exposed as a standalone module so that any subset of the workflow can be re-executed in isolation. The current state of completion of each stage is summarized in Table 1.

Table 1: Pipeline stages and current implementation status. Stage 5 is detailed in Section 3.

Stage	Description	Framework	Status
1	Random alloy configuration generation	Python	Complete
2	MD elastic-property evaluation	LAMMPS Python API	Complete
3	DFT reference calculations	Quantum Espresso + ASE	Complete
4	MEAM initialization and library merging	Python	Complete
5	Neural-network surrogate optimization	TensorFlow + LAMMPS	Planned

The data flow between stages is shown schematically in Figure 1. Stage 1 emits JSON descriptions of alloy configurations; Stage 2 simulates those configurations in LAMMPS and writes the measured elastic moduli to a shared results database. Stage 3 produces DFT reference values for every selected element and binary pair and stores them in a reference-data database. Stage 4 reads both of these and emits a merged, DFT-initialized pair of MEAM files. Stage 5 (planned) will consume those files together with the LAMMPS-evaluated (C_{11}, C_{12}) targets and produce a final, refitted MEAM parameter set. A divergent, composition-only neural-network branch (dashed in Figure 1) consumes the Stage 2 elastic database as a sanity check but never feeds back into the potential-fitting flow; it is described in Section 2.6.

What Stage 5 will do. Stage 5 will refit the cross-interaction terms of the merged MEAM library emitted by Stage 4 to better reproduce the LAMMPS-evaluated elastic targets. The plan, prototyped during this semester and detailed in Section 3, has four steps:

1. **Sample.** Draw N perturbed parameter vectors $\mathbf{p}_k = \mathbf{p}_0 \cdot (1+\boldsymbol{\eta}_k)$ around the Stage 4 baseline ($\eta_k \sim \mathcal{U}(-0.1, 0.1)$ on each binary (`Ec`, `re`, `alpha`)) and run LAMMPS over

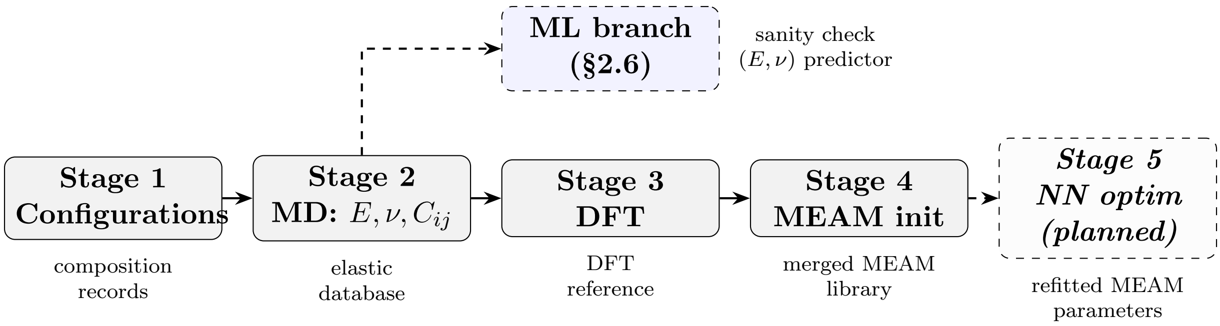

Figure 1: Five-stage pipeline architecture: Stages 1–4 (solid arrows) are complete, Stage 5 (dashed, italicised) is planned. The dashed ML side-branch above Stage 3 is the composition surrogate of Section 2.6, which consumes the Stage 2 elastic database as a sanity check but never feeds back into the potential-fitting flow.

every alloy in the elastic-property database to obtain $(C_{11}, C_{12})_k$ per composition. Samples whose majority of compositions fail a $C_{11} > 30\,\mathrm{GPa}$ physicality check are rejected.

2. **Train a surrogate.** Fit a feed-forward network $\mathbf{p} \mapsto (C_{11,k}, C_{12,k})_{k=1}^{N_{\text{entries}}}$ with a Huber loss. Predicting the elastic constants directly (rather than ν, whose ratio form is ill-conditioned in composition space) and using a robust loss are the two lessons carried over from the composition surrogate of Section 2.6.

3. **Inverse optimization.** Find $\hat{\mathbf{p}} = \arg\min_{\mathbf{p}} \|f_\theta(\mathbf{p}) - \mathbf{y}^\star\|^2$ by Adam descent on the normalised parameter vector, clipped to $\pm 2.5\sigma$ around the training mean to keep the optimization within the surrogate's domain of validity. A physical-stability penalty ($C_{11} + C_{12} > 0$, $C_{11} \geq C_{12}$, $\nu < 0.48$) will be added to keep the inverse optimizer inside the mechanically-stable region.

4. **Validate.** Re-evaluate $\hat{\mathbf{p}}$ in LAMMPS, derive $(E_{\text{opt}}, \nu_{\text{opt}})$ analytically from the predicted (C_{11}, C_{12}), and compare against a held-out set of named industrial alloys (Al 2024, 2219, 5052, 7075 plus extensions) and at least two non-aluminium alloys.

The supporting infrastructure (parallel LAMMPS sampling, MPI distribution, MEAM file I/O, C_{11}/C_{12} persistence in `results database`) is already in place from the Stage 2 work described in Section 2.3.

Stage 1 — Configuration Generation

The configuration generator produces randomized alloy compositions over the twelve elements covered by at least one of the MEAM library files currently available to the project: Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti and Zn. Each generated configuration is a JSON object containing the species composition (mass fractions summing to exactly unity at 10^{-4} precision), the simulation box edge in metres, the target temperature and the requested number of MD steps. A representative example, corresponding to a simplified Al 7075 composition, is shown below:

```
{
  "composition": { "Al": 0.91, "Mg": 0.029, "Zn": 0.061 },
  "box_size_m": 4e-09,
  "temperature": 0.1,
  "total_steps": 1000
}
```

To bias the dataset towards practically interesting alloys, aluminium is forced to be the dominant species whenever it is present in the random draw. Configurations corresponding to commercial alloys (Al 2024, Al 2219, Al 5052, Al 7075) are included as named reference entries in addition to the random ones. At the time of this report, 7924 configuration records have been generated, providing a wide population from which Stage 2 selects samples for MD simulation. The element-occurrence counts across this set are summarised in Figure 2: aluminium appears in $\sim 45\,\%$ of configurations (consistent with the Al-dominance bias), while every other element is sampled in roughly $30-37\,\%$ of records, giving the merge logic of Stage 4 enough cross-pair coverage to seed every binary interaction.

Stage 2 — MD Elastic Evaluation

Each configuration is loaded into LAMMPS (Thompson et al., 2022) through its Python API and its elastic response is computed with a six-point central-difference strain protocol.

Multi-element MEAM build of LAMMPS: The default LAMMPS distribution hard-codes the maximum number of MEAM species to five, which is too small for the twelve-element library generated in Stage 4. To enable the multi-element simulations required by this project, LAMMPS was rebuilt from source after raising the species ceiling in the MEAM headers to 20, and the resulting shared library was installed into the user's environment so that the Python API picks it up automatically. Without this step the

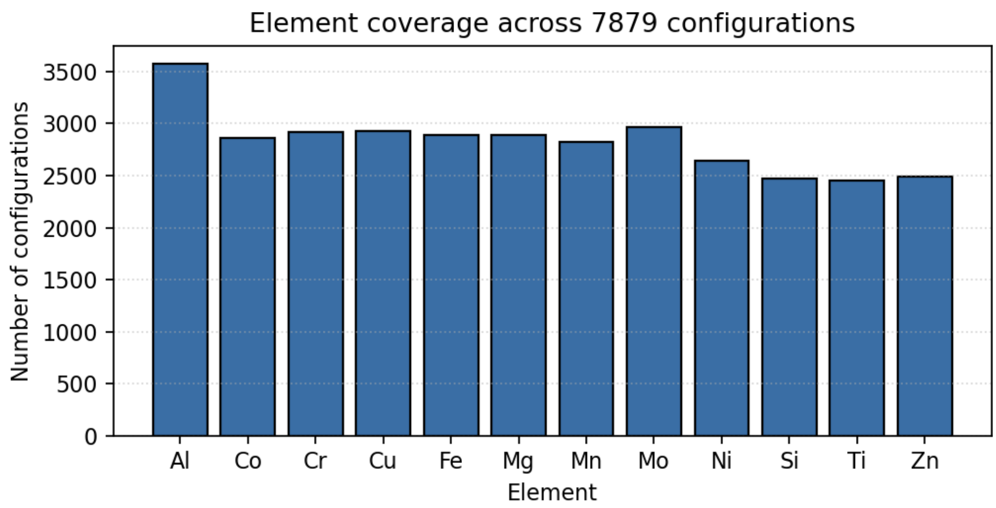

Figure 2: Number of configurations in which each element is present, out of the 7924 records currently in the dataset. The dominance of Al reflects the bias rule that retains aluminium as the majority species whenever it is drawn.

entire pipeline would be blocked on the first composition that contains more than five distinct elements.

Elastic-property protocol: The procedure is:

1. **Box construction** — a cubic supercell with edge of $\sim 4\,\mathrm{nm}$ is built using the dominant element's lattice structure and the composition-weighted lattice parameter. Atom types are then re-assigned by `set type/fraction` to match the requested composition.

2. **Relaxation** — an anisotropic pressure minimisation simultaneously relaxes both the cell shape and the atomic positions until the residual stress is below a fixed tolerance.

3. **Baseline stress recording** — the diagonal stress components σ_{xx}, σ_{yy} and σ_{zz} are recorded after a `run 0`.

4. **Strain protocol** — a strain $\delta = 10^{-3}$ is applied along each of x, y, z in turn, both positive and negative; the structure is re-minimized at each strain and the stress tensor is recorded.

5. **Elastic constant extraction** — C_{11} is computed from the axial-stress response and C_{12} from the transverse response by central differences, yielding three independent C_{11} samples and six independent C_{12} samples per simulation. The Voigt average (Liu et al., 2009) is used as the final estimate.

6. **Engineering moduli** — Young's modulus and Poisson's ratio are obtained from

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \qquad \nu = \frac{C_{12}}{C_{11} + C_{12}}. \tag{1}$$

Physicality filter. Configurations are discarded *and* removed from the configuration set if any of the following hold:

- $E < 0$ (negative Young's modulus);
- $\nu < 0$ (negative Poisson's ratio);
- $\nu \geq 0.48$ (close to the $(1 - 2\nu) = 0$ singularity);
- $C_{11} < C_{12}$ (Cauchy/mechanical-stability violation).

The "nearly 8000" figure quoted throughout this report therefore refers to the size of the dataset that has survived this quality filter, not to the number of configurations originally drawn.

Output and storage. All four quantities (E, ν, C_{11}, C_{12}) are persisted for every accepted configuration, so the composition surrogate of Section 2.6 (and the future Stage 5 parameter optimizer) can read them directly from the LAMMPS strain protocol described above.

Parallel evaluation. The elastic-only batch path dispatches configurations across worker processes, with one LAMMPS instance per physical core and OpenMP threading disabled to avoid hyperthread oversubscription. The sequential code path is preserved for the visualization and full-MD modes, where OVITO rendering and MD trajectories are inherently per-process. On a 4-physical-core / 8-logical-core host this gives roughly a 3–4× speedup over the sequential path.

The protocol has been executed on nearly 8000 randomly generated configurations; the resulting (E, ν) pairs are stored together with the composition in a shared elastic-property database.

Across the full sample set, E ranges from a few GPa for Mg/Zn-rich compositions to nearly 400 GPa for refractory-rich compositions. The dataset-level statistics, computed over the 7815 configurations that pass the physicality filter ($0 < E < 400$ GPa, $0 < \nu < 0.5$), are summarised in Table 2, and the corresponding distributions are shown in Figure 3. Figure 4 colours the E–ν scatter by the dominant element of each composition. These values constitute the elastic-property database consumed by the composition surrogate of Section 2.6.

Table 2: Summary statistics of the LAMMPS-evaluated elastic properties over the physicality-filtered dataset ($0 < E < 400$ GPa, $0 < \nu < 0.5$). Auto-generated from 7879 configuration records.

Quantity	N	Mean	Median	Std.	Min	Max
E (GPa)	7815	132.877	103.610	69.098	3.070	398.310
ν	7815	0.319	0.326	0.063	0.011	0.497

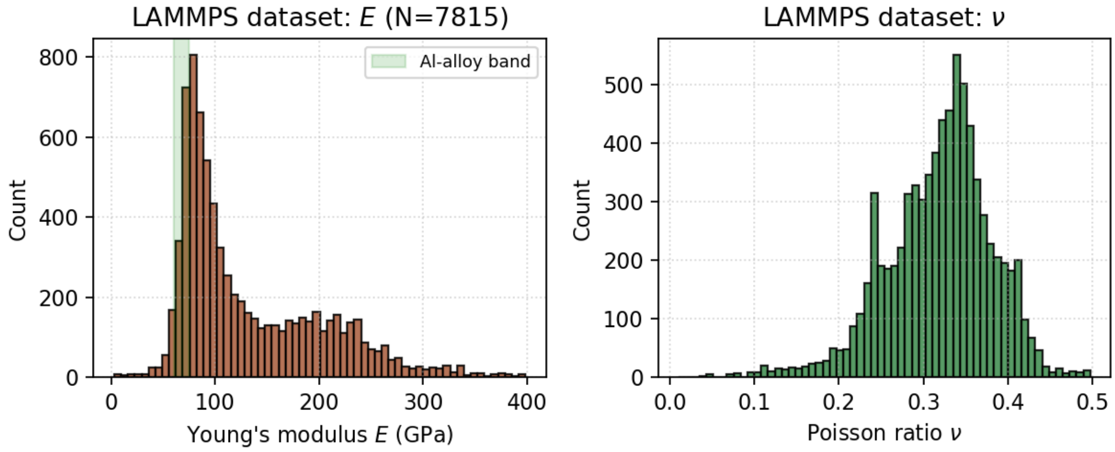

Figure 3: Histograms of E (left) and ν (right) across the LAMMPS-evaluated dataset. The shaded green band on the left panel marks the experimentally expected 60–75 GPa range for aluminium alloys, which contains 14.0 % of the physical sample.

Stage 3 — DFT Reference Calculations

The DFT reference stage uses Quantum Espresso (Giannozzi et al., 2009) (version 7.5, compiled locally with gfortran 13.3, OpenMPI 4.1 and FFTW3 3.3) driven through the Atomic Simulation Environment (Larsen et al., 2017). Before any production calculations were carried out, the QE–ASE coupling was validated on bulk FCC aluminium with a three-point sanity check (single-point SCF energy in the expected -540 to -530 eV window, residual atomic forces below 0.01 eV/Å, and diagonal-only stress tensor), all of which pass at the chosen plane-wave cut-offs. For each requested element the following quantities are then computed:

1. the equilibrium lattice constant a_0, cohesive energy E_{coh} and bulk modulus B from a seven-point equation-of-state scan fit to a Birch–Murnaghan equation (Angel, 2000);
2. an isolated-atom reference energy in a large empty box, used to convert the bulk total energy into a true cohesive energy;
3. the cubic elastic constants C_{11} and C_{12} from a stress–strain calculation analogous to the LAMMPS protocol of Section 2.3.

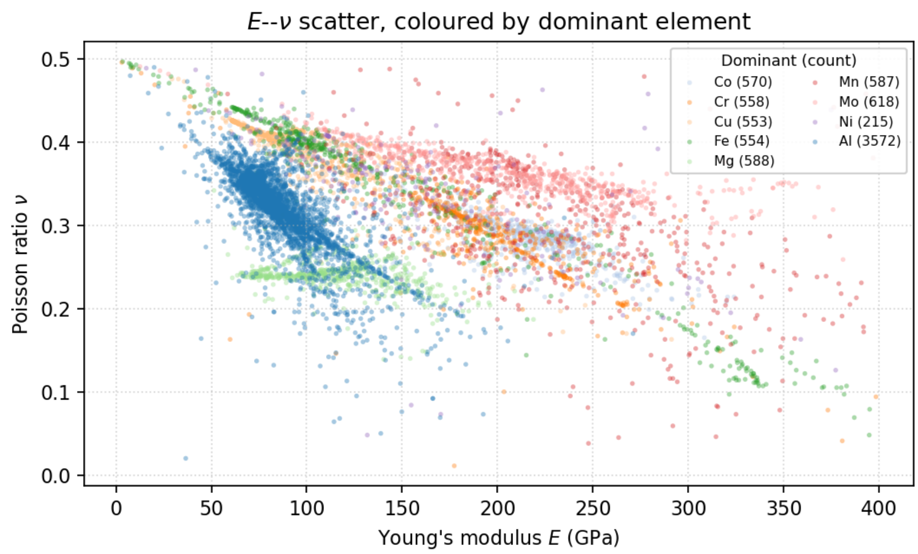

Figure 4: E–ν scatter of the LAMMPS-evaluated dataset, coloured by the dominant element of each composition. The Al-dominant cloud occupies the experimentally expected 60–75 GPa band; refractory-rich (Mo, Cr, Fe) compositions populate the high-E wing.

For each unordered binary pair (i, j) a single SCF calculation is run on the $L1_2$ or B2 reference structure (chosen based on the lattice types of the two elements) to obtain the formation energy $E_{\mathrm{form}}(i, j)$.

The plane-wave cut-off is set to $E_{\mathrm{cut}} = 40\,\mathrm{Ry}$ with a charge density cut-off of 320 Ry, Marzari–Vanderbilt smearing of width 0.02 Ry, and a $6 \times 6 \times 6$ Monkhorst–Pack mesh. Spin-polarised calculations are activated automatically for the magnetic elements Fe, Cr and Mn. Pseudopotentials are downloaded from the official Quantum Espresso pseudopotential library on demand; the EOS strain points and the binary-pair SCF calculations are dispatched to a process pool, which made it possible to obtain the full reference set on a single workstation.

The current DFT reference dataset covers ten elements (Al, Cu, Si, Ti, Zn, Cr, Fe, Mg, Mn and Mo) and 40 binary pairs. The fitted elemental properties for a representative subset are listed in Table 3. The Mn entry was retained at its tabulated value because the EOS scan failed to converge for the simplified α-Mn approximation, and the corresponding cross-terms therefore inherit a higher uncertainty. The fitted EOS curves are shown in Figure 5.

The 40 binary formation energies $E_{\mathrm{form}}(i, j)$ obtained from the $L1_2$/B2 SCF references are summarised as a symmetric heatmap in Figure 6. Negative values (blue) indicate exothermic mixing, expected for compatible *sd*/*sp* pairs such as Al–Ti or Cu–Ti, while strongly positive values (red) flag elements with low mutual solubility (notably the Mg/Zn pair against most transition metals). These signed values feed Eq. (2) of Stage 4 directly

Table 3: Elemental reference data extracted from DFT. E_{coh} is in eV/atom and B, C_{ij} in GPa.

Element	Lattice	a_0 (Å)	E_{coh}	B	C_{11}	C_{12}
Al	FCC	4.047	3.609	77.500	167.100	32.600
Cu	FCC	3.639	3.858	138.500	135.100	127.500
Si	DIAMOND	5.471	5.130	89.300	171.280	48.310
Ti	HCP	2.915	6.288	114.500	161.647	90.926
Zn	HCP	2.766	0.964	68.400	123.120	41.040
Cr	BCC	2.848	8.666	213.100	450.100	91.300
Fe	BCC	2.806	6.885	420.500	694.314	283.593
Mg	HCP	3.177	1.483	47.800	78.926	32.237
Mn†	BCC	8.910	3.000	70.000	113.077	48.462
Mo	BCC	3.169	10.353	280.800	407.600	63.200

† EOS scan did not converge; tabulated values retained, donor-library entries used at merge time.

and therefore propagate, unmodified, to the cross-term $E_c(i,j)$ entries of the merged MEAM library.

Stage 4 — MEAM Initialization and Library Merging

The DFT reference set is mapped onto MEAM parameters through analytic relations. The single-element mappings are direct:

$$\texttt{esub} \leftarrow E_{\mathrm{coh}},$$
$$\texttt{alat} \leftarrow a_0,$$

while the Rose shape parameter is set from the universal equation of state (Vinet et al., 1989):

$$\alpha = \sqrt{\frac{9\,B\,\Omega}{E_{\mathrm{coh}}}}, \qquad \Omega = \begin{cases} a_0^3/4 & \text{FCC}, \\ a_0^3/2 & \text{BCC}, \\ a_0^3\sqrt{2} & \text{HCP (approx.)}, \\ a_0^3/8 & \text{diamond}, \end{cases} \tag{2}$$

where B is converted from GPa to eV/Å^3 before substitution. As a worked example, for aluminium ($a_0 = 4.047$ Å, $E_{\mathrm{coh}} = 3.609$ eV, $B = 77.500$ GPa, FCC), Eq. (2) gives $\Omega = 16.572$ Å^3, $B_{\mathrm{eV/Å^3}} = 0.484$, and therefore $\alpha = \sqrt{9 \times 0.484 \times 16.572/3.609} \approx 4.471$.

The cross-interaction terms are seeded from the binary formation energies and the ele-

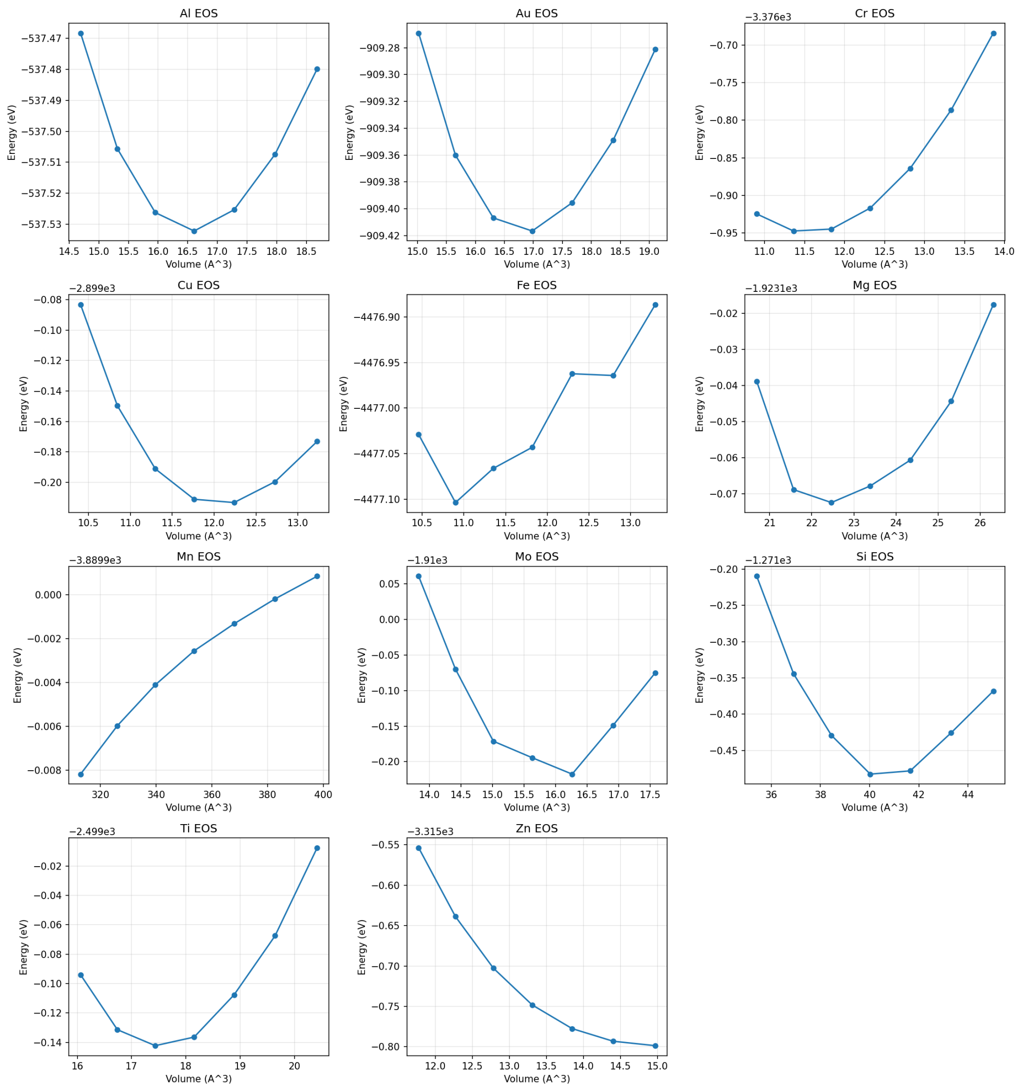

Figure 5: Birch–Murnaghan fits to the seven-point DFT equation-of-state scans for the elements computed by Stage 3. The fitted minimum determines a_0 and E_{coh}; the curvature determines B.

mental nearest-neighbour distances:

$$E_c(i,j) = \tfrac{1}{2}\left[\mathtt{esub}_i + \mathtt{esub}_j\right] + E_{\mathrm{form}}(i,j), \tag{3}$$

$$r_e(i,j) = \tfrac{1}{2}\left[d_{\mathrm{nn}}(i) + d_{\mathrm{nn}}(j)\right], \tag{4}$$

$$\alpha(i,j) = \tfrac{1}{2}\left[\alpha_i + \alpha_j\right]. \tag{5}$$

Automatic library discovery and merging. Because the existing public MEAM libraries each cover a different subset of elements, Stage 4 begins with a discovery and merge pass. The procedure is:

1. enumerate every available MEAM library file;

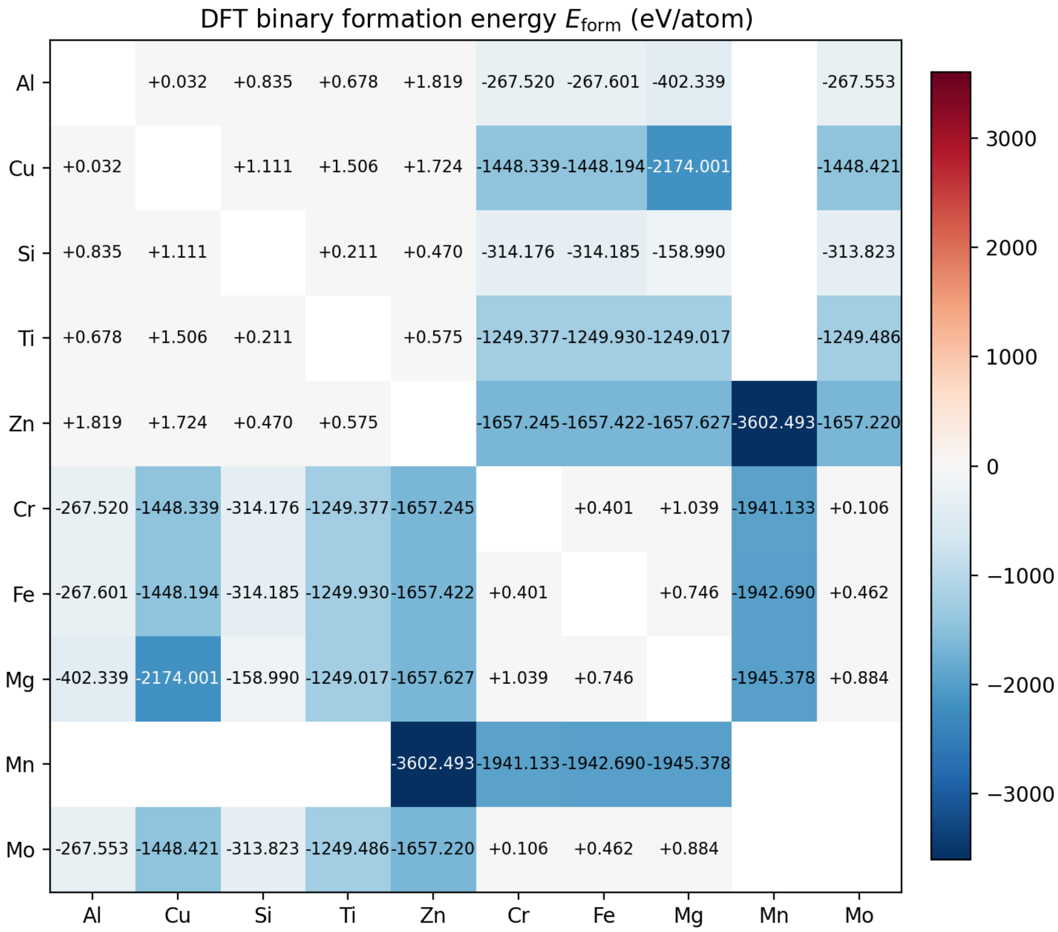

Figure 6: DFT binary formation energy $E_{\mathrm{form}}(i, j)$ in eV/atom for the ten elements with a converged elemental reference. The matrix is symmetric by construction. Cells corresponding to (self,self) entries are not computed and are left blank.

2. compute the smallest set of files whose union covers the requested element list;
3. parse the single-element entries from each chosen file and concatenate them, taking only the first occurrence of any duplicated species;
4. fill in any cross-interaction entry that is absent from the merged set with the value computed from the DFT reference;
5. write the resulting unified library and parameter files to a DFT-initialized output location.

The default twelve-element selection produces a single merged library covering Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti and Zn. As an illustration of the merge logic, an Al–Cu–Zn–Mg–Fe–Cr–Ti potential draws Al, Cu, Fe, Cr and Ti from the Fe–Mn–Ni–Ti–Cu–Cr–Co–Al library, Mg and Zn from the Al–Mg–Zn library, and synthesises the missing cross-terms (Cr–Mg, Ti–Zn, etc.) from the DFT formation energies.

The single-element parameters obtained by applying Eqs. (2) to the DFT reference are listed in Table 4. The Mn row is flagged because the EOS scan failed to converge for the simplified α-Mn approximation (Section 2.4); its α value is therefore retained only as a placeholder and is overwritten by the donor library entry during the merge step. All other rows feed directly into the `esub`, `alat` and `alpha` entries of the DFT-initialized parameter file.

Table 4: MEAM single-element initialization values computed from the DFT reference of Stage 3 by Eq. (2).

Element	Lattice	a_0 (Å)	E_{coh} (eV)	Ω (Å^3)	α
Al	FCC	4.047	3.609	16.569	4.471
Cu	FCC	3.639	3.858	12.051	4.930
Si	DIAMOND	5.471	5.130	20.464	4.473
Ti	HCP	2.915	6.288	35.022	5.985
Zn	HCP	2.766	0.964	29.931	10.920
Cr	BCC	2.848	8.666	11.549	3.994
Fe	BCC	2.806	6.885	11.046	6.156
Mg	HCP	3.177	1.483	45.357	9.063
Mn†	BCC	8.910	3.000	353.674	21.531
Mo	BCC	3.169	10.353	15.911	4.924

† EOS scan did not converge; values overwritten by the donor library at merge time.

Auxiliary Composition Surrogate (MD sanity-check branch)

This subsection describes the divergent ML branch shown as the dashed side-arrow in Figure 1. Unlike Stages 3–4, which seed the main pipeline that will eventually be closed by neural-network parameter optimization, the present branch consumes the Stage 2 MD output directly and never feeds back into the potential-fitting flow. Its sole purpose is to act as a fast sanity check on the consistency of the LAMMPS-generated dataset — if a composition-only network cannot recover even an approximate trend in (E, ν) from the dataset, the MD output itself is suspect — and to provide a ready-to-use predictor for novel compositions.

Each input vector has 10 entries (one mass fraction per element in {Al, Co, Cr, Cu, Fe, Mg, Mn, Ni, Ti, Zn}); the network has the architecture $10 \to 32 \to 20 \to 2$ with ReLU activations and is trained under a 70/30 train/validation split with the Adam optimizer.

Targets and Huber loss. The regression targets are the LAMMPS-evaluated (E, ν) pairs produced by the Stage 2 strain protocol described in Section 2.3. Records with $\nu \geq 0.48$ are dropped to remove configurations that lie close to the mechanical-stability bound and that destabilise the regression in composition space. The training loss is **Huber** ($\delta = 1$ in scaled units) rather than MSE, which bounds the gradient contribution of outlier samples (Mg/Zn-rich compositions whose LAMMPS runs occasionally yield extreme moduli).

Stopping gate. Training stops as soon as the *mean* relative error on the derived (E, ν) over the held-out 30 % validation set falls within `MAX_ERROR_PCT` = 20 % for both quantities, i.e. $\overline{|\Delta E|/E} \leq 20\,\%$ *and* $\overline{|\Delta\nu|/\nu} \leq 20\,\%$. This replaces an earlier per-sample gate that required *every* validation alloy to be within 20 %; that criterion was dominated by a small number of Mg/Zn-rich outliers and tended to drive the optimiser into false restarts just to satisfy the worst sample. The mean-based gate targets the bulk behaviour of the surrogate while still leaving the individual residuals visible in Figures 9–10; per-sample pass/fail counts at the same 20 % threshold are still emitted as diagnostics.

Table 5: Performance of the composition surrogate (N=7839; train=5487, val=2352).

Split / Target	MAE	RMSE	Std	Bias	R^2	r	MAPE
Train, C_{11} (GPa)	17.229	58.630	58.633	0.419	0.800	0.894	7.986 %
Train, C_{12} (GPa)	12.190	53.922	53.920	-0.895	0.694	0.834	16.334 %
Train, E (GPa, derived)	18.495	30.051	29.947	2.528	0.840	0.918	15.073 %
Train, ν (derived)	0.025	0.037	0.037	0.000	0.686	0.829	10.418 %
Val., C_{11} (GPa)	19.174	54.260	54.269	0.520	0.747	0.874	8.662 %
Val., C_{12} (GPa)	13.747	47.993	48.000	-0.580	0.554	0.767	14.725 %
Val., E (GPa, derived)	19.832	33.500	33.430	2.268	0.789	0.891	15.975 %
Val., ν (derived)	0.026	0.039	0.039	0.001	0.616	0.785	10.099 %

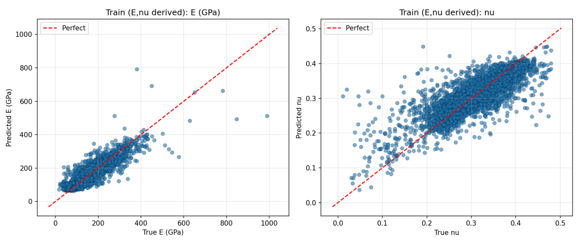

Figure 7: Training-set parity for the predicted E and ν.

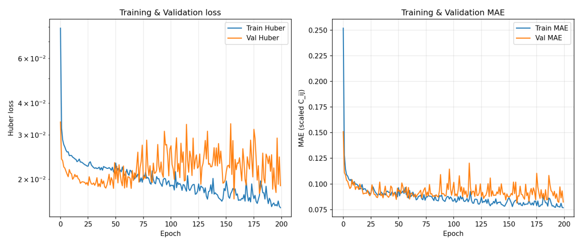

Figure 8: Training history of the surrogate: Huber loss and MAE on the scaled (E, ν) targets.

Future Outlook

Outstanding Tasks

The pipeline architecture, the DFT reference, the MEAM merge logic, and a composition-only sanity-check surrogate are all in place. The remaining work concentrates on closing the loop with a parameter-level neural-network optimization stage and on formal validation:

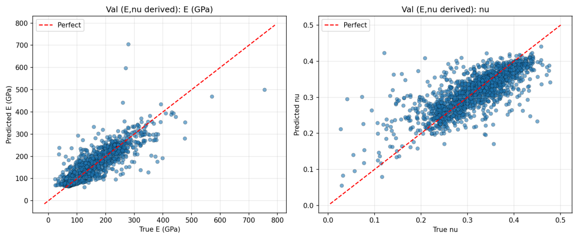

Figure 9: Held-out validation parity for (E, ν).

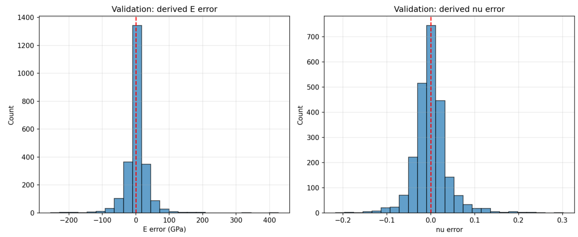

Figure 10: Validation error distributions for E and ν. Both distributions are approximately zero-centred, indicating no systematic bias; residual tails are dominated by Mg/Zn-rich compositions.

1. **Implement the Stage 5 parameter optimization.** The neural-network surrogate that maps MEAM parameters to alloy elastic properties — and the inverse-optimization step that uses it to refit the Stage 4 library against LAMMPS-evaluated targets — is the headline remaining task. The supporting infrastructure (parallel LAMMPS sampling, MPI distribution, MEAM file I/O) is already in place; what remains is to design a stable surrogate architecture, choose a robust loss, and add an explicit physical-stability penalty ($C_{11} + C_{12} > 0$, $C_{11} \geq C_{12}$, $\nu < 0.48$) that keeps the inverse optimizer inside the surrogate's domain of validity.

2. **Improve the DFT reference for Mn.** The α-Mn EOS scan does not converge and the Mn entry is a tabulated fallback. As an interim measure a helper backfills any missing or unphysical C_{ij} from the recorded B and a per-element Poisson-ratio default, so Stage 4 receives a complete and self-consistent set; this also rescues Si, Ti, Zn and Mg whose non-cubic lattices skip the elastic-constant computation entirely. A proper Mn calculation (simpler reference structure or different smearing scheme) is still planned.

3. **Build a validation set of named alloys.** A formal benchmark against industrial aluminium alloys (Al 2024, Al 2219, Al 5052, Al 7075, Al 6061, Al 6063, Al 5083) and at least two non-aluminium alloys (a steel and a Ti alloy) will be assembled once the Stage 5 optimizer starts producing physical results. Both LAMMPS-with-merged-MEAM baselines and final-optimized predictions will be reported.

4. **Write the final report and prepare the poster.** Sections from the present interim report will be carried over to the final report, with the Stage 5 results and the formal validation added. The report tooling has been automated: every figure and LaTeX table in the present document is regenerated automatically from the pipeline outputs, so refreshed LAMMPS or DFT data flow through to the report on the next compile.

The infrastructure required to execute these tasks at scale is already in place. The pipeline has built-in checkpoint/resume support: each of the DFT and MEAM-initialization stages writes its output to a deterministic location and the orchestrator detects which of those outputs already exist, so a re-run after a failure or a parameter change skips work that has already been completed.

Updated Semester Plan

The Gantt chart of Figure 11 reflects the work that has already been completed and the schedule of the remaining tasks. The first five work packages from the proposal (literature review, DFT reference calculations, MEAM initialization, NN surrogate training, and the first pass at inverse optimization) are now complete; validation, scaling, and report/poster preparation occupy the rest of the semester.

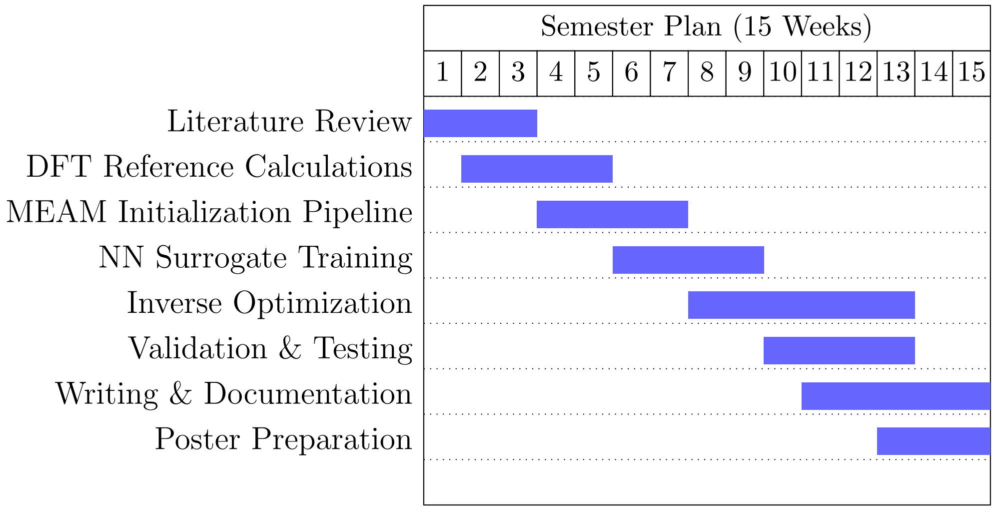

Figure 11: Semester plan with the actual progress to date. Items 1–4 are complete; items 5–8 are the focus of the second half of the semester.

Description of Work Packages

- **Literature Review (W1–W3).** Survey of MEAM, EAM and neural-network potential literature. Completed.
- **DFT Reference Calculations (W2–W5).** Build of Quantum Espresso, pseudopotential management and the EOS/elastic-constant/binary-pair pipeline. Completed.
- **MEAM Initialization Pipeline (W4–W7).** Auto-discovery and merging of available MEAM library files, DFT-to-MEAM analytic mapping, generation of the unified twelve-element library. Completed.

- **NN Surrogate Training (W6–W9).** Sampling of the MEAM parameter space, parallel LAMMPS evaluation and training of the parameter–property surrogate. First pass complete; will be re-run with a larger sample set.
- **Inverse Optimization (W8–W13).** Gradient-based back-propagation through the surrogate to fit elastic targets. First pass complete with the limitations described above; ongoing.
- **Validation & Testing (W10–W13).** Comparison against named industrial alloys and external validation sets. Begun.
- **Writing & Documentation (W11–W15).** Final report and code documentation.
- **Poster Preparation (W13–W15).** Final poster for the end-of-semester presentation.

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